Online Appendix to “Optimal Disclosure Windows”

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OA.1 Benchmark: Perfectly Informed Agent

As a benchmark, consider the scenario in which the agent is informed of whether the state is $\theta = 0$ or $\theta = 1$ at time 0. As disclosure starts later, the stopping game converges to the benchmark case where the agent is perfectly informed. As mentioned in Proposition 3, the waiting time in the continuation stopping game is bounded above by $\overline{w}^*$, which is the waiting time in this benchmark. Proposition OA.1 formalizes this argument (the characterization of $\overline{w}^*$ is relegated to the Online Appendix).

Proposition OA.1. *In the continuation game, there exists* $0 < \overline{w}^* < \infty$ *such that the informed agent randomizes over stopping times in* $[0, \overline{w}^*]$ *and uninformed agent stops with probability* 1 *at* $\overline{w}^*$.

The proof is the same as the proof of Theorem 1.B (ii) as is thus omitted.

OA.2 Omitted Proofs for Section 3

OA.2.1 Proof of Lemma 19

Take the derivative of $V^*(t)$ with respect to t,

$$V^{*\prime}(t) = \frac{e^{\lambda t - r w^*(t)}\lambda\mu(1-\mu) - e^{\lambda t - r w^*(t)}\mu\left(r(1-\mu) + e^{\lambda(t+w^*(t))}(r+\lambda)\mu\right) w^{*\prime}(t)}{\left(1 + \left(e^{\lambda(t+w^*(t))} - 1\right)\mu\right)^2}.$$

It suffices to show

$$e^{\lambda t - r w^*(t)}\lambda\mu(1-\mu) - e^{\lambda t - r w^*(t)}\mu\left(r(1-\mu) + e^{\lambda(t+w^*(t))}(r+\lambda)\mu\right) w^{*\prime}(t) < 0,$$

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which is equivalent to showing

$$w^{*\prime}(t) > \frac{\lambda(1-\mu)}{r(1-\mu)+e^{\lambda(t+w^*(t))}(r+\lambda)\mu}.$$

Recall that

$$w^{*\prime}(t) = \frac{\beta e^{(\beta+r)w^*(t)}+r}{\beta\left(e^{(\beta+r)w^*(t)}-1\right)}\frac{\lambda(1-\mu)}{r(1-\mu)+e^{\lambda(t+w^*(t))}(r+\lambda)\mu}.$$

The first term is greater than 1 and the second term is positive. The result follows.

OA.2.2 Proof of Lemma 20

I prove the case for the uninformed agent, the informed agent follows from the same algebra. By definition, $Y(t|t_0)$ is given by

$$\begin{aligned} Y(t|t_0) =& (1-\rho(t_0))\int_{t_0}^{t} \lambda e^{-\lambda(s-t_0)}e^{-r(s-t_0)}V^*(s)\mathrm{d}s \\ &+\left((1-\rho(t_0))e^{-\lambda(t-t_0)}+\rho(t_0)\right)e^{-r(t-t_0)}U^*(t). \end{aligned}$$

Take the derivative with respect to t,

$$\begin{aligned} Y'(t|t_0) =& e^{-r(t-t_0)}\frac{e^{r(t-t_0)}\rho(t)}{(1-\rho(t_0))e^{-\lambda(t-t_0)}+\rho(t_0)} \\ &\cdot\left(\rho'(t)\left(V^*(t)-U^*(t)\right)+\rho(t)\left(U^{*\prime}(t)-rU^*(t)\right)\right). \end{aligned}$$

where the first line is strictly positive and the second line is proportional to $y(t)$. The result follows.

OA.2.3 Proof of Lemma 21

Take the derivative of $Y(t)$ with respect to t. After simplifying, the derivative is proportional to

$$\lambda(1-\mu)e^{-\lambda t}\left(V^*(t)-U^*(t)\right)+\left((1-\mu)e^{-\lambda t}+\mu\right)\left(U^{*\prime}(t)-rU^*(t)\right).$$

Recall that the derivative of the best-case scenario payoff $Y_0(t)$ is $\rho'(t)-r\rho(t)$, which, by assumption, is negative. Because $V^*(t)-U^*(t)<0$, that is, the informed agent's equilibrium continuation payoff is lower than the uninformed, it suffices to show if $\rho'(t)-r\rho(t)<0$ for all t, $U^{*\prime}(t)-rU^*(t)<0$ for all t.

The rest of the proof proceeds by brute force calculations. One can explicitly write down the expression for $U^{*\prime}(t)-rU^*(t)$. After simplifying,

$$
\begin{aligned}
U^{*\prime}(t) - rU^*(t) = & \frac{e^{-(\beta+r)w^*(t)}}{(\beta-\lambda)\left(\rho(t)\left(e^{\lambda w^*(t)}-1\right)+1\right)} r\rho(t)\left(\lambda\rho(t)e^{\lambda w^*(t)} - e^{\beta w^*(t)}(\beta-\lambda+\lambda\rho(t))\right) \\
& + \rho'(t)\frac{e^{-(\beta+r)w^*(t)}}{(\beta-\lambda)\left(\rho(t)\left(e^{\lambda w^*(t)}-1\right)+1\right)^2} \\
& \qquad \cdot \left(\lambda\rho(t)\left(e^{\beta w^*(t)} - e^{\lambda w^*(t)}\right)\left(1+\rho(t)e^{\lambda w^*(t)} + 1 - \rho(t)\right) + (\beta-\lambda)e^{\beta w^*(t)}\right) \\
& + w^{*\prime}(t)\frac{e^{-rw^*(t)}\rho(t)}{(\beta-\lambda)\left(\rho(t)\left(e^{\lambda w^*(t)}-1\right)+1\right)^2} \\
& \qquad \cdot \Bigg(r(\lambda-\beta) + \rho(t)\left(r(\beta-2\lambda+\lambda\rho(t))\right) + \rho(t)\lambda e^{(\lambda-\beta)w^*(t)}(\beta-\lambda+r) \\
& \qquad + \lambda e^{(\lambda-\beta)w^*(t)}\rho(t)^2\left(\lambda-\beta-r-(\lambda+r)e^{\beta w^*(t)} + re^{\lambda w^*(t)} + \beta e^{\lambda w^*(t)}\right) \\
& \qquad + e^{(\lambda-\beta)w^*(t)}\rho(t)(\lambda-\beta)(\lambda+r)e^{\beta w^*(t)}\Bigg)
\end{aligned}
$$

The expression for $w^{*\prime}(t)$ is given by (23). It can be verified that for each t, whenever $\rho'(t) - r\rho(t) < 0$, the right-hand side is negative for all w^*. So with the equilibrium waiting time $w^*(t)$, $U^{*\prime}(t) - rU^*(t)$.

OA.3 Proofs for Section 5

OA.3.1 Proof of Proposition 4

Let $q_D(w)$ denote the decision maker's belief that $\theta = 1$ conditional on no signals arriving in $[0, w]$. By Bayes' rule,

$$
q_D(w) = \frac{\eta}{\eta + e^{-\lambda w}(1-\eta)}. \tag{OA.1}
$$

Given this belief $q_D(w)$, from the perspective of the beginning of the continuation stopping game, denote the decision maker's expected payoff from stopping at w by $D(w)$, then

$$
\begin{aligned}
D(w) = & (1-\eta)\int_0^w e^{-rs}\lambda e^{-\lambda s}e^{-\beta s}\mathrm{d}s \\
& + (1-\eta)\int_0^w e^{-rs}e^{-\lambda s}\beta e^{-\beta s}\left(1-(q_D(s)-0)^2\right)\mathrm{d}s \\
& + (1-\eta)e^{-rw}e^{-\lambda w}e^{-\beta w}\left(1-(q_D(w)-0)^2\right) \\
& + \eta\int_0^w e^{-rs}\beta e^{-\beta s}\left(1-(q_D(s)-1)^2\right)\mathrm{d}s \\
& + \eta e^{-rw}e^{-\beta w}\left(\left(1-(q_D(w)-1)^2\right)\right).
\end{aligned} \tag{OA.2}
$$

Take the derivative of $D(w)$ with respect to w. After simplifying,

$$D'(w) \propto \lambda(1-q_D(w))q_D(w)^2 - r\left(1-(1-q_D(w))q_D(w)\right). \tag{OA.3}$$

Define

$$R(w;\eta) := \left(\frac{e^{-\lambda w}(1-\eta)+\eta}{e^{-\lambda w}(1-\eta)} + \left(\frac{\eta + e^{-\lambda w}(1-\eta)}{\eta}\right)^2\right)^{-1}. \tag{OA.4}$$

Rearrange the right-hand side of (OA.3) and substitute (OA.1) for $q_D(w)$, $D'(w) \le 0$ if and only if $R(w;\eta) \le r/\lambda$.

Claim OA.1. For any w and η, $\partial R(w;\eta)/\partial w$ and $\partial R(w;\eta)/\partial \eta$ have the same sign.

Proof. To facilitate the proof, note that $q_D(w)$ depends on both w and η. With a slight abuse of notation, denote (OA.1) by $q_D(w;\eta)$. Then $R(w;\eta)$, defined in (OA.5), can be written as

$$R(w;\eta) = \left(\frac{1}{1-q_D(w;\eta)} + \frac{1}{q_D(w;\eta)^2}\right)^{-1}. \tag{OA.5}$$

Then
$$\frac{\partial R(w;\eta)}{\partial w} = \frac{\partial R(w;\eta)}{\partial q_D}\frac{\partial q_D(w;\eta)}{\partial w} \text{ and } \frac{\partial R(w;\eta)}{\partial \eta} = \frac{\partial R(w;\eta)}{\partial q_D}\frac{\partial q_D(w;\eta)}{\partial \eta}.$$

By (OA.1), $\partial q_D(w;\eta)/\partial w > 0$ and $\partial q_D(w;\eta)/\partial \eta > 0$. The result follows. □

Claim OA.2. $R(w;\eta)$ has the following properties.

(i) For all $0 < \eta < 1$, $R(w;\eta)$ is single-peaked in w: there exists a unique $w_R^*(\eta) = \arg\max_w R(w;\eta)$ with $w_R^*(\eta) \ge 0$ such that $R(w;\eta)$ is strictly increasing in w for $w \le w_R^*(\eta)$ and strictly decreasing in w for $w > w_R^*(\eta)$.

(ii) There exists a unique $0 < \eta_R < 1$ such that

(a) for all $\eta < \eta_R$, $w_R^*(\eta) > 0$, $w_R^{*\prime}(\eta) < 0$, and $R(w_R^*(\eta);\eta)$ is constant in η;

(b) for all $\eta \ge \eta_R$, $w_R^*(\eta) = 0$ and $R(w_R^*(\eta);\eta) = R(0;\eta)$ is decreasing in η;

(c) Moreover, $w_R^*(\eta)$ is continuous for all $0 < \eta < 1$. That is, $\lim_{\eta\to\eta_R} w_R^*(\eta) \to 0$.

(iii) For all $0 < \eta < 1$, $\lim_{w\to\infty} R(w;\eta) \to 0$.

Proof. These properties follow directly from operating on (OA.4).

Specifically, for (ii) (a) and (c), if $w_R^*(\eta)$ is interior, namely, $w_R^*(\eta) > 0$, $w_R^*(\eta)$ has a closed form which can be obtained by solving $\partial R(w;\eta)/\partial w = 0$. It follows from the functional form of $w_R^*(\eta)$ that $w_R^{*\prime}(\eta) < 0$ and $\lim_{\eta\to\eta_R} w_R^*(\eta) \to 0$. By definition,

$$\frac{\mathrm{d}R(w_R^*(\eta);\eta)}{\mathrm{d}\eta} = \frac{\partial R(w_R^*(\eta);\eta)}{\partial \eta} + \frac{\partial R(w_R^*(\eta);\eta)}{\partial w}\frac{\mathrm{d}w_R^*(\eta)}{\mathrm{d}\eta}.$$

The second term on the right-hand side is zero because $\partial R(w_R^*(\eta);\eta)/\partial w = 0$. By Claim OA.1, the first term $\partial R(w_R^*(\eta);\eta)/\partial\eta = 0$. Therefore, $\mathrm{d}R(w_R^*(\eta);\eta)/\mathrm{d}\eta = 0$, which means $R(w_R^*(\eta);\eta)$ is constant in η.

For (ii) (c), it follows from the definition (OA.4) that $R(0;\eta) = \left(\frac{1}{1-\eta} + \frac{1}{\eta^2}\right)^{-1}$ and is decreasing in η. $\square$

Part (i). It follows from Claim OA.2 (ii) that $R(0;\eta_R) = \max_{\eta\in(0,1)} R(w_R^*(\eta);\eta)$. Denote this maximum value by

$$\Delta := R(0;\eta_R).$$

Then for all $w \geq 0$ and $0 < \eta < 1$, $R(w;\eta) \leq R(0;\eta_R) = \Delta$. This means if $\Delta \leq r/\lambda$, $R(w;\eta) \leq \Delta$, which means $D'(w) < 0$. (In fact, for any parameters, Δ is a constant number is approximately 0.1916.)

Part (ii). Fix $r/\lambda < \Delta$. The shape of $D(w)$ depends on η. There are three cases.

Case 1. By Claim OA.2, for all $\eta \geq \eta_R$, $R(w;\eta)$ is decreasing in w for all $w \geq 0$ and is thus maximized at $w = 0$. Because $R(0;\eta)$ is decreasing in η, there exists $\overline{\eta} > \eta_R$ such that $R(0;\overline{\eta}) = r/\lambda$ and $R(0;\eta) < r/\lambda$ (and therefore $R(w;\eta) < r/\lambda$) for all $\eta > \overline{\eta}$. This implies $D'(w) \leq 0$ for all $w \geq 0$ and is thus maximized at $w = 0$ for all $\eta \geq \overline{\eta}$.

Case 2. By Claim OA.2, for all $\eta < \eta_R$, $R(w;\eta)$ is single-peaked in w and is maximized at $w_R^*(\eta) > 0$. Because $R(0;\eta)$ is decreasing in η, there exists $\tilde{\eta} < \eta_R$ such that there exists a unique $w_{DM}^*(\eta)$ where $R(w_{DM}^*(\eta);\eta) = r/\lambda$, $R(w;\eta) < r/\lambda$ for $w < w_{DM}^*(\eta)$, and $R(w;\eta) > r/\lambda$ for $w > w_{DM}^*(\eta)$. This implies $D(w)$ single-peaked in w is uniquely maximized at $w = w_{DM}^*(\eta) > 0$ for $\eta \in [\tilde{\eta}, \overline{\eta})$.

Case 3. For all $\eta < \tilde{\eta}$, $R(0;\eta) < r/\lambda$ and $R(w;\eta)$ intersects with the line r/λ twice. Denote these two intersections by $w_L(\eta)$ and $w_H(\eta)$ with $w_L(\eta) < w_H(\eta)$. That is, $R(w_L(\eta);\eta) = R(w_H(\eta);\eta) = r/\lambda$.

Moreover, $R(w;\eta) < r/\lambda$ for $w < w_L(\eta)$ and $w \geq w_H(\eta)$, and $R(w;\eta) > r/\lambda$ for $w \in [w_L(\eta), w_H(\eta))$. This means that $D(w)$ decreases for $w \in [0, w_L(\eta))$, increases for $w \in [w_L(\eta), w_H(\eta))$, and decreases for $w \geq w_H(\eta)$. Therefore, for each $\eta < \tilde{\eta}$, $D(w)$ has two local maxima, one at $w = 0$ and one at $w = w_H(\eta)$. To determine which is the global maximizer, one can obtain $D(w_H(\eta))$ by plugging in the closed form expression for $w_H(\eta)$, and compare it with $D(0) = 1 - (1-\eta)\eta$. It can be shown that there exists $\underline{\eta} < \tilde{\eta}$ such that $D(w_H(\eta)) < 1 - (1-\eta)\eta$ for all $\eta < \underline{\eta}$, and $D(w_H(\eta)) \geq 1 - (1-\eta)\eta$ for all $\eta \in [\underline{\eta}, \tilde{\eta})$. Then the optimal waiting time $w_{DM}^*(\eta) = 0$ for $\eta < \underline{\eta}$ and $w_{DM}^*(\eta) = w_H(\eta)$ for $\eta \geq \underline{\eta}$.

This conclude the proof. As an illustration, Figure OA.1 plots the function $D(w)$ for several values of η. The resulting optimal waiting time $w^*_{DM}(\eta)$ is given by ??.

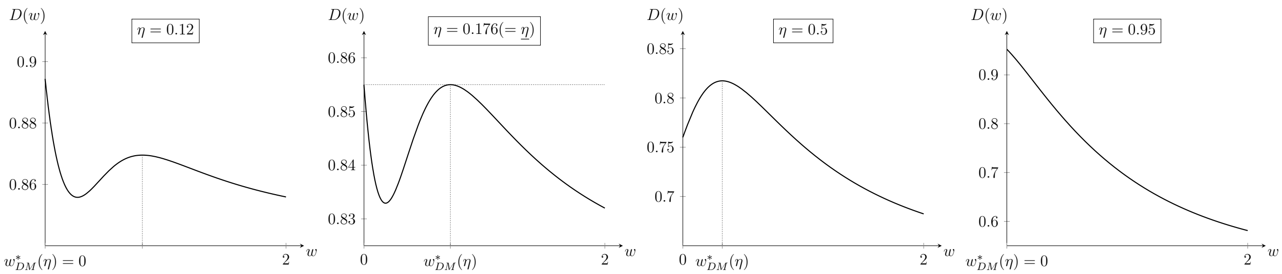

Figure OA.1: Decision maker's expected payoff from stopping at w $D(w)$ for $r = 0.5, \lambda = 5$, and $\beta = 0.5$. From left to right, the values of η are respectively $\eta = 0.12, 0.176, 0.5, 0.95$.